

Ignacio Francisco Barraza Rioja

Backend Software Engineer

Chile · Remote · Open to Relocation

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PROFESSIONAL SUMMARY

Backend Software Engineer with **3+ years of experience** building and maintaining production systems with **Node.js, TypeScript, NestJS, and PostgreSQL**. Experienced in designing REST APIs, optimizing databases and application performance, implementing automated testing, and building maintainable backend architectures using Clean and Hexagonal Architecture. Delivered production systems supporting **700+ users**, reduced average API response times by **30%**, and automated workflows that reduced operational reporting effort by approximately **70%**. English: C1 – Professional Working Proficiency.

EXPERIENCE

Full Stack Developer · MapVx

Feb 2023 – Present

Node.js · NestJS · TypeScript · PostgreSQL · TypeORM · Docker · Jest · n8n

- Worked as a **full stack developer**, participating in backend architecture decisions, API design, and database optimization while contributing to frontend features with Angular when needed.
- Developed, maintained, and implemented **Angular** components.
- Optimized **PostgreSQL** schemas and **TypeORM** queries, reducing average API response time by **30%**.
- Implemented caching strategies that improved perceived application performance by approximately **40%** across two core applications.
- Built an automated ticket reporting workflow integrating **Notion, n8n, and Jira**, reducing management and reporting time by approximately **70%**.
- Designed and developed backend services using **Node.js, NestJS, Express, and TypeScript**, applying modular architecture and Clean Code principles.
- Containerized applications using **Docker**, standardizing development and deployment environments.
- Developed and maintained automated **unit tests with Jest**, improving regression detection and production reliability.
- Acted as primary technical owner for bug reports and feature requests from Customer Success, translating issues into production fixes.
- Estimated, planned, and tracked development work using **Jira** within Agile/Scrum workflows.

Intern, Backend Developer · MapVx

Jan 2025 – Feb 2025

Node.js · Express · TypeScript · PostgreSQL · TypeORM · Hexagonal Architecture

- Developed a RESTful backend for the company's client portal using **Node.js, Express, TypeScript, PostgreSQL, and TypeORM**.
- Implemented **hexagonal architecture** (Ports & Adapters), ensuring clean domain separation, maintainability, and scalability.
- Designed and optimized database queries, handling relational data, arrays, and date operations efficiently.
- Built support for **multilanguage content management**, enabling flexible and localized user experiences.
- Applied best practices in request validation and API design to deliver robust and reliable endpoints.

Teaching Assistant – Computer Science · Universidad de Tarapacá

Mar 2024 – Dec 2024

Iquique, CL

- Mentored **50+ students** in Object-Oriented Programming (OOP) and relational database design using SQL.
- Facilitated workshops on software engineering best practices, focusing on logic optimization and **Clean Code** principles.

KEY PROJECTS

Hyōsei – Academic Evaluation Platform

- Production platform used across **4 official university events**, supporting **700+ active users** across multiple concurrent events.
- Achieved average API response times of approximately **45 ms** under typical workloads, with **zero downtime** throughout all supported events.

- Backend built with **Node.js, NestJS, TypeORM, PostgreSQL, and JWT**; frontend with **Angular and Tailwind CSS**. Deployed via **Railway** and **Vercel**.
- Applied **hexagonal architecture** with role-based access control (Admin, Evaluator, Visitor), Zod validation, and secure JWT + HttpOnly cookie authentication.
- Currently migrating backend from Express to **NestJS** to improve modularity, maintainability, and separation of concerns.

Kōbun – Cross-Platform PDF Utility

- Designed and delivered a production-ready cross-platform desktop application for PDF page extraction using **Python, PySide6, and PyMuPDF**.
- Implemented a layered architecture inspired by **Domain-Driven Design**, separating domain, application, infrastructure, and presentation layers.
- Developed **439 automated tests** including property-oriented tests and regression coverage, with the domain suite running independently of external frameworks.
- Built cross-platform packaging for **Linux and Windows** (.deb + Windows installer) with automated **CI/CD via GitHub Actions**: semantic versioning, changelog generation, matrix builds, and release automation.

TECHNICAL SKILLS

Backend: Node.js, TypeScript, NestJS, Express, REST APIs, PostgreSQL, TypeORM, MongoDB

Architecture & Quality: Clean Architecture, Hexagonal Architecture, SOLID, DDD, Unit Testing, Jest, Zod

Cloud & Infrastructure: Docker, GitHub Actions, CI/CD, Railway, Vercel, GCP Cloud Run

Event-Driven & Automation: Webhooks, n8n, Event-Driven Architecture, RabbitMQ, Kafka

Frontend: Angular, RxJS, React, Next.js (App Router), Tailwind CSS

Additional: Java, Spring Boot, AWS Lambda, SQS, SNS, EventBridge, Terraform, Vitest, Playwright, Git, Agile/Scrum

LANGUAGES

Spanish: Native · **English:** C1 – Professional Working Proficiency · **Japanese:** Basic (in progress)

EDUCATION

Civil Engineering in Computer Science

Universidad de Tarapacá – Chile · Thesis Grade: 6.8 / 7.0

Degree awarded: January 2026